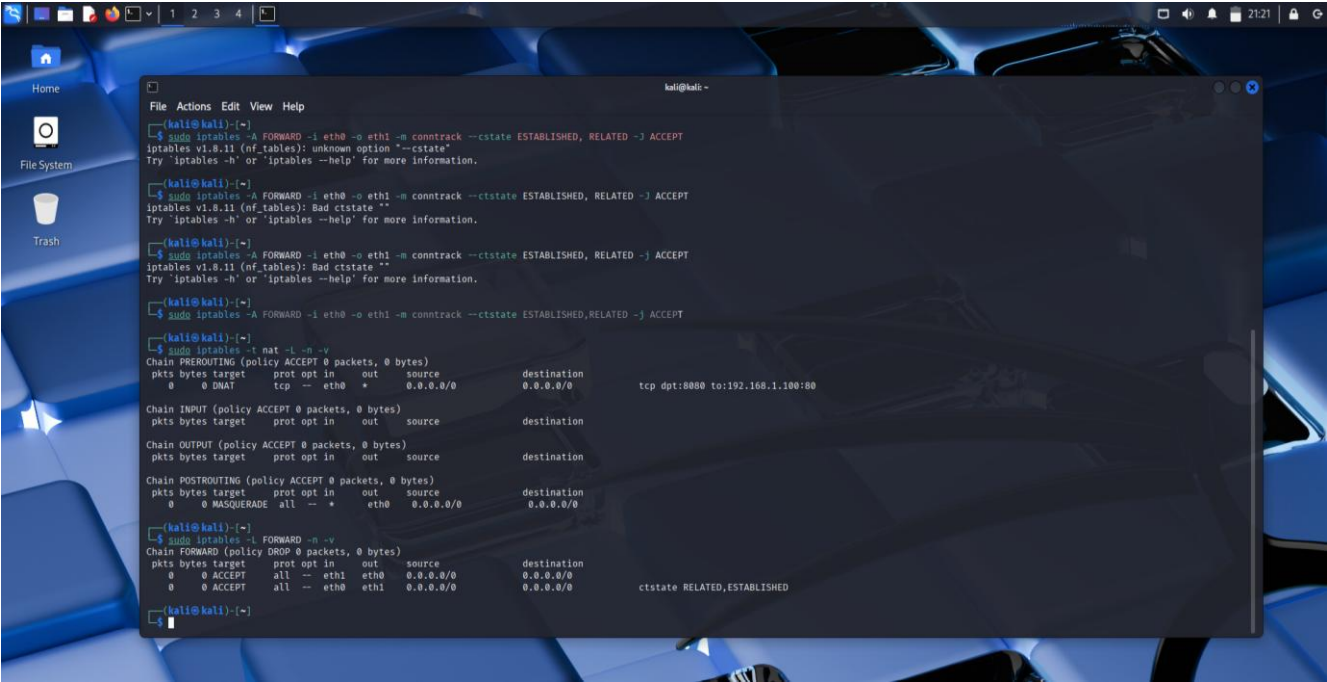


Lab Activity: Advanced Firewall Routing and NAT

Step 1: Table Inspection distinguish between the filter table and the nat table:

- Check NAT: `sudo iptables -t nat -L -n -v`
- Check Filters: `sudo iptables -L FORWARD -n -v`



Step 2: Critical Thinking & Security Analysis

1. Obfuscation: How does MASQUERADE protect the Confidentiality of the internal network's size and structure?

- **Answer:** MASQUERADE (a form of Source NAT) translates all outgoing traffic from your internal network so that it looks like it is coming from the Gateway's single public IP address. Because of this, anyone looking from the outside only sees one device (the Gateway). They cannot see the private IP addresses, they cannot count how many devices are connected, and they cannot map out how your internal network is organized.

2. Port Remapping: We mapped Public Port 8080 to Internal Port 80. What is the security advantage of using a non-standard port on the public-facing interface?

- **Answer:** Using a non-standard port (like 8080) provides a defense layer known as "security through obscurity." Automated malicious bots and scripts constantly scan the internet looking for open default ports (like port 80 for HTTP or 22 for SSH) to attack. By putting your public-facing service on a high, unexpected port, you hide it from most casual, automated scans and significantly reduce the amount of malicious traffic hitting your server.

3. The "Missing Link": If you successfully configure Part 2 (SNAT) but forget Part 1 (IP Forwarding), will the internal clients have internet? Why or why not?

- **Answer:** No, the internal clients will not have internet access. By default, a Linux machine acts as an "End Host," meaning it will immediately drop any network packets that aren't specifically addressed to itself. Enabling IP Forwarding is what turns the Linux kernel into a "Transit Host" (a router). If you forget to turn it on, the kernel will simply drop the internal network's outgoing packets before they ever make it to your iptables NAT rules.

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